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Acados-based Implementation of Nonlinear MPC for Magnetic Levitation

Project thesis in Cybernetics and Robotics
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ABSTRACT

Magnetic Levitation (MagLev) systems present demanding real time control challenges, as they are characterized by fast, inherently unstable, and highly non-linear dynamics. Stabilizing these systems requires control solutions capable of meeting exceptionally tight control deadlines, imposing strict demands on computational efficiency. This thesis investigates the feasibility and efficacy of advanced Model Predictive Control (MPC) strategies, powered by the high-performance optimization framework `acados`. The goal is to investigate how to perform an MPC implementation through the `acados` framework, evaluate its benefits and limitations, and investigate whether it is suited for controlling a complex magnetic levitation system.

The research primarily focuses on a simplified 1-Degree Of Freedom (DOF) levitating ball system, which serves as a benchmark for deploying computationally efficient Nonlinear MPC. Three distinct MPC architectures are implemented and evaluated: a Standard MPC controller, a Stochastic MPC controller for handling parametric uncertainties, and a Tube-based MPC controller for guaranteed robustness.

Crucially, the computational analysis demonstrates that the `acados`-based solvers are substantially more efficient than comparable standard MPC implementations, achieving solution times in the millisecond range. Specifically, for the standard MPC formulation, `acados` achieves a speedup factor of 22.3× for short prediction horizons ($N = 5$) and scaling to a factor 7.6× for longer horizons ($N = 40$). This significant speedup is achieved without sacrificing substantial control performance, as all three `acados`-based controllers demonstrate comparable tracking accuracy and stability to the benchmark MPC. The achieved control performance and computational efficiency lays the groundwork for deploying a robust, `acados`-based MPC solution on a complex MagLev platform.

CONTENTS

Abstract **i**

Contents **ii**

List of Figures **iv**

List of Tables **v**

Abbreviations **vi**

1 Introduction **1**
1.1 Background 1
1.2 Previous work 1
1.3 Objectives 2
1.4 Sustainable Development Goals 2
1.5 Declaration of Generative AI 2
1.6 Limitations 2
1.7 Outline of The Thesis 3

2 Collaboration **4**
2.1 Reinforcement Learning 4
2.2 `acados` 5
2.3 Model Predictive Control 5
2.4 Estimator/Observer 5

3 Theory **6**
3.1 Optimal Control Problems (Optimal Control Problem (OCP)s) . . . 6
3.1.1 Mathematical Formulation 6
3.1.2 Modeling Constraints 7
3.1.3 Methods for Solving an OCP 7
3.2 Model Predictive Control 8
3.2.1 The Receding Horizon Principle 8
3.2.2 MPC Objective Function 8
3.2.3 System Integration and Architecture 9
3.2.4 Advanced MPC Formulations 9

4 Methods **11**
4.1 Physical Systems Under Consideration 11
4.1.1 Levitating Ball 11
4.2 Numerical Optimization through `acados` 12
4.2.1 What is `acados`? 12
4.2.2 Why Do We Use It? 12
4.3 Design of Experiments (DoE): Central Composite Design (CCD) . . 13
4.3.1 Face-Centered Central Composite Design (FCCD) 14
4.4 Performance Metrics 14
4.5 Experiments 14
4.5.1 Experiment 1: Computational Efficiency Benchmarking . . . 15
4.5.2 Experiment 2: Trajectory Tracking Performance 15
4.5.3 Experiment 3: Parameter Sensitivity and Optimization via Response Surface Methodology (RSM) 17
4.5.4 Experiment 4: Robustness Analysis under Model Mismatch . 18

5 Results **19**
5.1 Experiment 1: Computational Efficiency Benchmarking 19
5.2 Experiment 2: Trajectory Tracking Performance 21
5.3 Experiment 3: Parameter Sensitivity and Optimization via Response Surface Methodology (RSM) 24
5.4 Experiment 4: Robustness Analysis under Model Mismatch 28

6 Discussion **34**
6.1 Interpretation of results 34
6.1.1 Experiment 1: Computational Efficiency Benchmarking . . . 34
6.1.2 Experiment 2: Trajectory Tracking Performance 35
6.1.3 Experiment 3: Parameter Sensitivity and Optimization via RSM 36
6.1.4 Experiment 4: Robustness Analysis under Model Mismatch . 38
6.2 Future work 39
6.2.1 Short-term 39
6.2.2 Medium-term 39
6.2.3 Long-term 39

7 Conclusion **41**

References **42**

LIST OF FIGURES

3.1 Relationship between controller, system and estimator. 9

4.1 Visualization of the three components of a CCD. 13

5.1 Computational performance comparison for varying horizons N. . . 20
5.2 Comparison of MPC-controllers for sigmoid trajectories with slopes 5, 10, and 15. 22
5.3 Comparison of MPC-controllers for sinusoidal trajectories with angular frequencies 0.1, 1, and 10. 23
5.4 Comparison of `acados` MPC, `CasADi` MPC, and RL performance on sigmoid trajectory with slope 10. 24
5.5 Contour plot showing the state weights q_h and q_v against Root Mean Square Error (RMSE) for slope 10 26
5.6 Contour plot showing the state weights q_h and q_v against RMSE for slope 15 26
5.7 Contour plot showing the state weights q_h and q_v against the average RMSE 28
5.8 Model mismatch with lighter ball for sigmoid trajectories with slopes 5, 10, and 15. 30
5.9 Model mismatch with lighter ball for sinusoidal trajectories with angular frequencies 0.1, 1, and 10. 31
5.10 Model mismatch with heavier ball for sigmoid trajectories with slopes 5, 10, and 15. 32
5.11 Model mismatch with heavier ball for sinusoidal trajectories with angular frequencies 0.1, 1, and 10. 33

LIST OF TABLES

4.1 Physical Parameters of the 1-DOF Environment 12
4.2 Parameters for the Sigmoid Reference Trajectory 16
4.3 Parameters for the Sinusoidal Reference Trajectory 16
4.4 Tuning Parameters for the CCD Optimization. 18

5.1 Computational Time Comparison: `acados` vs. `CasADi` (ms) 20
5.2 RMSE for various MPC-controllers for sigmoid trajectories. 21
5.3 RMSE for various MPC-controllers for sinusoidal trajectories. . . . 21
5.4 RMSE comparison of `acados` MPC, `CasADi` MPC, and Reinforcement Learning (RL) on sigmoid trajectory with slope 10. 21
5.5 `CasADi` MPC parameters. 24
5.6 Central Composite Design (CCD) levels for MPC Tuning. 25
5.7 ANOVA breakdown of factor importance for MPC RMSE for slope 10. 25
5.8 ANOVA breakdown of factor importance for MPC RMSE for slope 15. 27
5.9 RMSE with lighter ball for sigmoid trajectories. 29
5.10 RMSE with lighter ball for sinusoidal trajectories. 29
5.11 RMSE with heavier ball for sigmoid trajectories. 29
5.12 RMSE with heavier ball for sinusoidal trajectories. 29

ABBREVIATIONS

List of all abbreviations in alphabetic order:

AI Artificial Intelligence

ANOVA Analysis of Variance

CCD Central Composite Design

DoE Design of Experiments

DOF Degree Of Freedom

EKF Extended Kalman Filter

LLM Large Language Model

LQR Linear Quadratic Regulator

MagLev Magnetic Levitation

MHE Moving Horizon Estimator

MPC Model Predictive Control

NLP Non Linear Program

NMPC Nonlinear MPC

OCP Optimal Control Problem

OVAT One Variable At a Time

PPO Proxial Policy Optimization

QP Quadratic Program

RL Reinforcement Learning

RMSE Root Mean Square Error

RSM Response Surface Methodology

SCQP Sequential Convex Quadratic Programming

SQP Sequential Quadratic Programming

CHAPTER

ONE

INTRODUCTION

1.1 Background

Magnetic levitation is a technology that utilizes magnetic fields to lift objects without any physical support other than the magnetic force itself. By counteracting the gravitational pull with electromagnetic or electrodynamic forces, the suspended object remains in a state of stable equilibrium. As noted by [1], the fundamental advantage of all MagLev applications is the total absence of physical contact between the moving part and its guide or base. This elimination of mechanical contact effectively removes friction and wear, leading to higher energy efficiency, reduced maintenance requirements, and an extended operational lifespan for the system components.

While high-speed transportation systems are the most prominent example of this technology, MagLev is increasingly employed across a diverse range of engineering sectors. In industrial applications, magnetic bearings are utilized in high-efficiency pumps, compressors, and energy-storage flywheels to allow for near-frictionless rotation. Furthermore, the technology plays a critical role in specialized fields such as biomedical engineering for heart pumps, aerospace for launch assistance systems, and even in civil engineering for advanced elevator designs [1]. This versatility demonstrates that the principles of magnetic levitation are not merely limited to transportation but serve as a robust solution for any precision system requiring non-contact motion control.

This thesis is a part of a larger project regarding a MagLev platform called "Maggy". This project revolves around a small take-home laboratory which is currently under development at the Norwegian University of Science and Technology (NTNU). More detailed information about the take-home lab can be found on GitHub [2].

1.2 Previous work

The GitHub repository for the Maggy project contains all the work done on the project since 2022. This includes detailed instructions for example usage of the

physical system, student contributions in form of project and master's theses, and a `MATLAB` simulator for the system made by Hans Alvar Engmark. Two important works for this thesis are Mikael Mjelde's master's thesis about robust MPC [3], and Hans Alvar Engmark and Kiet Tuan Hoangs article *Modeling and Control of a Magnetic Levitation Platform* [4], which explains the complex dynamics of the Maggy platform.

1.3 Objectives

The primary objectives of this thesis are:

1. Implementation of 3 different MPC formulations using the `acados` framework: Standard MPC, Stochastic MPC and Tube MPC
2. Evaluation and comparison of the real-time feasibility of the controllers above against benchmark `CasADi` MPC controllers.
3. Investigation of the trade-off between computational efficiency and tracking performance.

1.4 Sustainable Development Goals

This project contributes to the following of UN's Sustainable Development Goals:

- **Goal 4: Quality Education**: The thesis is a part of a take-home lab that enables educational accessibility to advanced control strategies.
- **Goal 9: Industry, Innovation and Infrastructure**: The thesis is about MagLev systems, which is a field of research that contains many applications that improves the industry and infrastructure. One prominent example is the high-speed MagLev transportation systems.

1.5 Declaration of Generative AI

This work utilized Consensus.app for literature research and the Large Language Model (LLM) Gemini for assistance with programming and linguistic refinement. While Artificial Intelligence (AI) tools suggested specific phrasings and provided code structures, all technical content and academic arguments were independently assessed, edited, and finalized by the author. AI was not used to answer the assignment in its entirety. The author assumes sole responsibility for the academic content of the assignment.

1.6 Limitations

The primary limitation of the study is that it utilizes a 1-DOF levitating ball system as a benchmark. While this serves as a necessary proof of concept, it remains a significant simplification of the full MagLev platform. The 1-DOF model does not fully capture the nonlinearities and the inherently less stable dynamics present in the three-dimensional Maggy system.

Secondly, due to time constraints, the implementation of the Tube and Stochastic MPC variants was not finalized to the same degree as the Standard MPC. Specifically, these robust formulations do not currently include constraints on the maximum change in control force between sampling intervals. Because the other controllers utilize this constraint to ensure smoother actuation, the resulting tracking error for the Tube and Stochastic variants is not directly comparable to the Standard MPC. This discrepancy limits the ability to draw definitive conclusions regarding the tracking precision of these advanced controllers relative to the standard implementation.

1.7 Outline of The Thesis

The remainder thesis is divided into the following chapters:

- **Collaboration**: Description of the collaboration towards the Maggy project.
- **Theory**: Provides the background information relevant to the project.
- **Methods**: Description of the systems considered, introduction to `acados` and experimental methodology.
- **Results**: Experimental results.
- **Discussion**: Discussion on results and future work.
- **Conclusion**: Summary of the main findings and future work.

CHAPTER

TWO

COLLABORATION

During this semester, the project regarding the *Take home maglev lab - Maggy* was conducted as a collaborative effort by four students: Eivind Bruaset, Thomas Aleksander Jonsson, Linus Edvardsen, and Marius Jullum Faanes. While we shared a common experimental platform, each member focused on a distinct research domain. This section presents the authors, as well as their specific domain of research for the project.

Linus Edvardsen: Reinforcement Learning (RL).

Eivind Bruaset: The `acados` library and solver implementation.

Marius J. Faanes: Model Predictive Control (MPC) formulation.

Thomas A. Jonsson: The estimator/observer design for the system.

Although we are submitting separate reports, the work heavily relies on the interoperability of these domains. To facilitate this, we held weekly supervision meetings with Prof. Damiano Varagnolo and conducted joint software development sessions. Linus developed the initial iteration of a 1-DOF ball simulator, which we distributed via GitHub. Throughout the semester, we iteratively improved this simulator, integrating our respective control and estimation strategies to ensure a unified development environment.

Note: This section has been written collaboratively with equally shared authorship and appears in all four reports.

2.1 Reinforcement Learning

Instead of using a classical controller to control the Maggy, the objective of this domain was to determine if a neural network could be trained to generate the correct input and stabilize the system using a model-free approach. This work utilized the Proxial Policy Optimization (PPO) algorithm to learn control policies directly from interaction with the simulator. The research established a development pipeline starting with a simplified 1-DOF levitating ball environment to tune

hyperparameters, before scaling to the full 6-DOF MagLev system. A key part of the collaboration involved benchmarking these RL agents against the Model Predictive Control (MPC) standard, specifically comparing tracking accuracy and real-time computational feasibility across various network architectures.

2.2 `acados`

To address the strict real-time requirements of the *Maggy* platform, this domain investigates the `acados` framework as a high-performance alternative to traditional optimization solvers. The objective is to determine if a solver utilizing tailored QP algorithms and optimized linear algebra routines can achieve the control performance and execution speed necessary for stabilizing inherently unstable dynamics. This work implements and evaluates three different MPC architectures using `acados`: standard MPC, Stochastic MPC, and Tube-based MPC. A central part of the project involves benchmarking these `acados`-based controllers against both model-free RL agents and MPC implementations using `CasADi` to compare tracking precision and computational efficiency.

2.3 Model Predictive Control

While the current simulation framework of the *Maggy* utilizes linearization and Linear Quadratic Regulator (LQR) control, this domain focuses on a higher-fidelity approach. Previous implementations of MPC on this system, specifically by Mikael Mjelde and Ørnulf Damsgaard in their master's theses, relied on linearizing the system dynamics around an equilibrium point. The primary goal for this project was to move beyond linearization and formulate the MPC as a full Non Linear Program (NLP). Furthermore, to specifically investigate strategies for improving system robustness under uncertainty, three distinct variants were implemented and compared: a standard MPC, a Tube-based MPC, and a Stochastic MPC.

2.4 Estimator/Observer

Similar to the MPC, the previous iterations of the Moving Horizon Estimator (MHE) for the *Maggy* was also implemented as a linear optimizer around a linearization point. In this project the MHE was to be developed fully non-linear, and with an accompanying Extended Kalman Filter (EKF) as an alternative non-linear observer for comparison. Because of the promising framework, the MHE was also to be implemented using `acados`. The overall goal was to implement the observers and find out how they behave under various environmental scenarios.

CHAPTER

THREE

THEORY

This chapter introduces the theoretical information relevant to the methods presented in the thesis. Firstly, the concept of optimal control problems is introduced. Secondly, the concept of Model Predictive Control (MPC) is presented. While the following sections provide the necessary context for this study, a more comprehensive treatment of these topics can be found in the literature cited within each subsection.

3.1 Optimal Control Problems (OCPs)

For complex dynamic systems, such as MagLev systems, the MPC feedback law is obtained by repeatedly solving OCPs [5]. To understand the concept of MPC, this section is dedicated to explaining OCPs. The content and notation in this section are mostly adapted from [6].

The primary objective of an optimization problem is to minimize (or maximize) a real-valued function that depends on a set of variables, which may be subject to specific constraints. An OCP is a specialized optimization problem where these variables represent a sequence of control inputs designed to drive a system along an optimal trajectory while satisfying physical and operational limits.

3.1.1 Mathematical Formulation

Following the notation established by Nocedal and Wright [6], a general constrained optimization problem is formulated as follows:

$$\min_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x}) \tag{3.1}$$

subject to:

$$c_i(\mathbf{x}) = 0, \quad i \in \mathcal{E}, \tag{3.2}$$

$$c_i(\mathbf{x}) \geq 0, \quad i \in \mathcal{I}, \tag{3.3}$$

where:

- $f(\mathbf{x})$ is the **objective function**, a smooth real-valued function that defines the performance criteria (e.g., minimizing tracking error).
- $\mathbf{x}$ represents the **decision variables**, which in an OCP context are the system states and control inputs over a finite horizon.
- $c_i(\mathbf{x})$ are the **constraint functions**.
- $\mathcal{E}$ and $\mathcal{I}$ are the finite sets of indices for **equality** and **inequality** constraints, respectively.

The **feasible set** Ω is defined as the collection of all points $\mathbf{x}$ that satisfy both the equality and inequality constraints.

3.1.2 Modeling Constraints

Constraints are essential for ensuring that the mathematical solution is physically implementable. In the context of dynamic systems, they are typically categorized into two types:

- **Equality Constraints ($c_i(\mathbf{x}) = 0, i \in \mathcal{E}$):** These represent the system dynamics. In an OCP, these equations ensure that the relationship between subsequent states follows the underlying physical model of the plant
- **Inequality Constraints ($c_i(\mathbf{x}) \geq 0, i \in \mathcal{I}$):** These represent physical boundaries, such as actuator limits (e.g., maximum applicable force) or physical constraints on the decision variable

3.1.3 Methods for Solving an OCP

Solving an OCP involves finding a point $\mathbf{x}^*$ in the feasible set such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all nearby feasible points. Because OCPs are often non-linear and constrained, they are typically solved using iterative algorithms that generate a sequence of estimates converging toward a solution. Common strategies described in Nocedal and Wright [6] include:

1. **Sequential Quadratic Programming (SQP):** This method models the problem by solving a quadratic programming subproblem at each iteration. It is widely used for systems with significant nonlinearities.
2. **Interior-Point Methods:** Also known as barrier methods, these use a barrier function to penalize points that approach the boundary of the feasible set, allowing the solver to treat the problem as a sequence of parameterized unconstrained problems.
3. **Active-Set Methods:** These algorithms focus on identifying which inequality constraints are "active" at the solution and treat them as equality constraints while ignoring the inactive ones during specific iterations.

3.2 Model Predictive Control

MPC is an optimization-based control strategy that utilizes a process model to forecast future behavior and calculate optimal control actions. The defining characteristic of MPC is its ability to explicitly integrate system constraints, such as actuator saturation or safety boundaries, directly into a dynamic optimization problem solved in real-time [5].

3.2.1 The Receding Horizon Principle

The operational core of MPC is the receding horizon strategy. At each sampling instant, the controller solves an open-loop OCP over a finite prediction horizon N. While the optimization produces a sequence of optimal control inputs, only the first element in this sequence is implemented. At the next time step, the horizon "recedes", a new state measurement is taken to provide feedback, and the optimization is repeated. This mechanism allows the controller to compensate for disturbances and model-plant mismatch [7].

The general execution of an MPC loop is formalized in Algorithm 1, adapted from the framework presented by [7].

Algorithm 1 Model Predictive Control Algorithm

```
Set the prediction horizon N and initialize the system.
for each sampling instant k = 0, 1, 2, ... do
    Measure or estimate the current state x_k of the plant.
    Solve a dynamic optimization problem on the prediction horizon from k to
    k + N, with x_k as the initial condition
    Apply the first optimal control action u*_k to the plant.
end for
```

3.2.2 MPC Objective Function

The objective function is the mathematical expression that the MPC algorithm seeks to minimize at each sampling instant. It quantifies the desired performance by penalizing deviations from reference trajectories and the control effort required to maintain them. In most practical applications, a quadratic cost function is used, as it results in a Quadratic Program (QP) problem that can be solved efficiently in real-time [7].

For a prediction horizon N, the objective function J is typically formulated as:

$$J = \sum_{k=0}^{N-1} \left(\|\mathbf{x}_{k+1} - \mathbf{x}_{ref}\|_{\mathbf{Q}}^2 + \|\mathbf{u}_k - \mathbf{u}_{ref}\|_{\mathbf{R}}^2 + \|\Delta \mathbf{u}_k\|_{\mathbf{R}_\Delta}^2 \right) \tag{3.4}$$

where the notation $\|\mathbf{a}\|_{\mathbf{W}}^2$ represents the weighted squared norm $\mathbf{a}^\top \mathbf{W} \mathbf{a}$. The performance of the controller is governed by the selection of the following weight matrices:

- **State Weighting Matrix (Q):** A positive semi-definite matrix that penalizes the tracking error. Larger values in $\mathbf{Q}$ encourage the controller to drive the system states toward the reference $\mathbf{x}_{ref}$ more aggressively.
- **Control Weighting Matrix (R):** A positive definite matrix that penalizes the magnitude of the control input $\mathbf{u}_k$. This matrix is used to limit energy consumption and prevent actuator saturation by penalizing large control actions.
- **Slew Rate Penalty ($\mathbf{R}_\Delta$):** This matrix penalizes the change in control input, defined as $\Delta\mathbf{u}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$. Penalizing the rate of change is critical in mechanical systems to prevent high-frequency "chatter" and reduce physical wear on the actuators [7].

The relative magnitudes of $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{R}_\Delta$ define the "tuning" of the MPC. For instance, increasing $\mathbf{R}$ relative to $\mathbf{Q}$ results in a more conservative controller that prioritizes smooth actuation over rapid setpoint tracking.

3.2.3 System Integration and Architecture

In a practical control hierarchy, the MPC does not operate in isolation. It functions as part of a closed-loop system that includes a state estimator and the physical plant. As shown in Figure 3.1, the estimator (e.g., a Kalman Filter or a Moving Horizon Estimator) processes sensor measurements $\mathbf{y}_k$ from the plant and and control output $\mathbf{u}_k$ to provide the MPC with an estimate of the current state $\hat{\mathbf{x}}_k$. The MPC then utilizes this estimate to compute the optimal input, effectively closing the feedback loop and ensuring robustness against unmeasured disturbances.

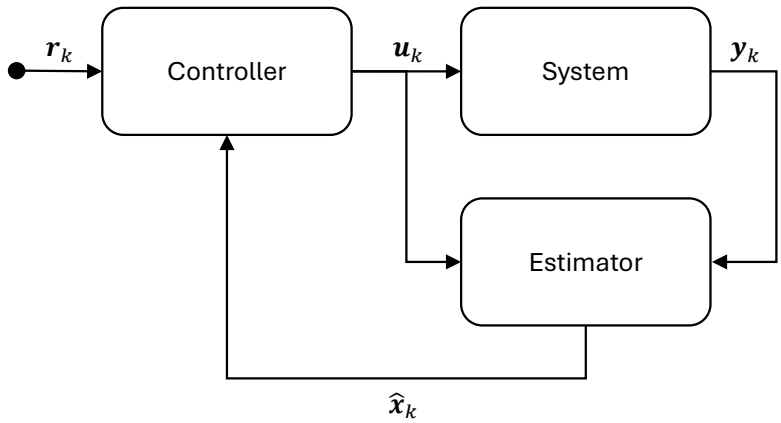

Figure 3.1: Relationship between controller, system and estimator.

3.2.4 Advanced MPC Formulations

Beyond the standard nominal MPC framework, specialized formulations exist to handle uncertainty more rigorously. Since the primary focus of this thesis is aimed at the standard MPC and the `acados` library, only a brief overview of Tube and Stochastic MPC is given in this subsection. As mentioned in Chapter 2, Marius' project studies advanced MPC formulations more rigorously.

3.2.4.1 Tube MPC

Tube MPC is a robust control technique designed to handle bounded disturbances and model uncertainties. As described by Feng et al. [8], this method decomposes the control task into two parts: a nominal MPC problem and an ancillary feedback controller. The nominal controller optimizes performance by planning a trajectory for a simplified, disturbance-free model.

To account for real-world uncertainties, the ancillary controller works to keep the actual system state within a bounded "tube" centered around this nominal path. By shrinking the state constraints to accommodate the size of this tube, the controller guarantees that the actual system satisfies the physical and operational limits despite the presence of disturbances. This separation of nominal planning and local regulation allows Tube MPC to provide rigorous robustness guarantees without the extreme computational burden of calculating worst-case scenarios for every possible state [8].

3.2.4.2 Stochastic MPC

Stochastic MPC addresses uncertainty by incorporating probabilistic models of disturbances rather than assuming worst-case bounds. As discussed in [9], traditional robust methods often lead to excessive conservatism by preparing for extreme scenarios that rarely occur. In contrast, Stochastic MPC exploits the statistical properties of uncertainty to allow for a more balanced control strategy.

Instead of enforcing hard boundaries, this approach utilizes chance constraints, which guarantee that the probability of a constraint violation remains below a specified threshold. This shift from a "worst-case" to a "risk-management" framework allows the controller to operate the system closer to its physical and economic limits, providing a quantifiable trade-off between aggressive performance and acceptable risk [9].

CHAPTER

FOUR

METHODS

4.1 Physical Systems Under Consideration

The original goal of the project was to implement MPC using `acados` on the existing Maggy simulator. However, due to time constraints, the work primarily focuses on the 1-DOF system. Therefore, this section only describes the 1-DOF system, while a full description of the Maggy system can be found in [4].

4.1.1 Levitating Ball

The 1-DOF levitating ball system simulates a point mass which can only move along the vertical axis. The mass is subject to a constant gravitational acceleration g downwards and an input force u. To simulate actuator saturation, the applicable force is constrained to $u \in \{-F_{max}, F_{max}\}$. The input force is defined such that a positive value for u denotes a positive force upwards. The resulting dynamics are given by Newtons Second Law:

$$\dot{v} = \frac{u}{m} - g \tag{4.1}$$

where v is the vertical velocity of the ball, and m is the ball's mass. Defining the height of the ball as h and the state vector as $\mathbf{x} = \begin{bmatrix} h & v \end{bmatrix}^\top$, results in the following continuous state space formulation:

$$\begin{bmatrix} \dot{h} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} h \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u + \begin{bmatrix} 0 \\ -g \end{bmatrix} \tag{4.2}$$

Moreover, to simulate that the force cannot change indefinitely fast, the change in u from one time step to the next (Δu) is constrained by $\pm \Delta u_{max}$. Finally, to simulate the ground, a hard physical constraint at height $h = 0$ is implemented, such that the velocity is set to zero in case of a collision. The parametric values used for the experiments are listed in Table 4.1.

Description	Parameter	Value	Unit
Ball Mass	m	1.0	kg
Gravitational Acceleration	g	9.81	m/s^2
Force Saturation Limits	F_{max}	± 50	N
Change in Force Limits	Δu_{max}	± 15	N/s
Ground Height	h_{ground}	0.0	m
Simulation Time Step	Δt	0.02	s

Table 4.1: Physical Parameters of the 1-DOF Environment

4.2 Numerical Optimization through `acados`

Model predictive control in real time embedded settings requires numerical tools that are both computationally efficient and flexible. `acados` is a high-performance open-source toolkit designed specifically for solving OCPs and implementing nonlinear model predictive control on platforms with strict timing constraints. It combines state of the art optimization algorithms with fast C implementations to enable real time feasible control of complex dynamic systems.

4.2.1 What is `acados`?

`acados` provides a modular collection of embedded optimization routines tailored to the structure of OCPs. Its architecture offers both flexibility and high computational performance, avoiding the typical trade-off encountered in many MPC tools based on code generation. The framework is written in C and relies internally on the BLASFEO linear algebra library to achieve near–hand-optimized speed for the small- to medium-scale matrices typical in embedded control applications [10].

In addition to its computational core, `acados` offers user-friendly interfaces in Python and MATLAB and is fully compatible with `CasADi`, enabling efficient symbolic modeling and automatic differentiation of nonlinear dynamics. This makes it possible to construct Nonlinear MPC (NMPC) controllers with nonlinear models, nonlinear constraints, and a wide range of cost formulations, all within a unified environment.

4.2.2 Why Do We Use It?

A key motivation for using `acados` in this project is its ability to deliver real time feasible NMPC, even when sampling times are on the order of milliseconds or below. The literature demonstrates that `acados` achieves sub-millisecond iteration times in closed-loop experiments on fast, unstable systems such as the inverted pendulum [11]. This level of performance is essential for controlling our physical Magnetic Levitation (MagLev) system, whose inherently unstable dynamics require rapid feedback to maintain levitation.

The design of `acados` explicitly targets embedded scenarios with tight timing requirements, enabled by:

- Exploiting optimal control structure through multiple shooting and tailored

QP formulations.

- Efficient Sequential Quadratic Programming (SQP)-type algorithms and real time iteration schemes.
- High-performance linear algebra through BLASFEO.
- Flexible Hessian approximations using Gauss-Newton and Sequential Convex Quadratic Programming (SCQP), providing a balance between speed and robustness.

These features ensure that the solver scales well with the prediction horizon and system dimension, making `acados` a uniquely suited high-performance solver for maintaining closed-loop stability in the MagLev system's highly demanding real time environment.

4.3 Design of Experiments (DoE): Central Composite Design (CCD)

When optimizing complex systems such as a NMPC controller, the primary objective is often to understand how various tuning parameters, such as the prediction horizon or weight matrices, influence the overall performance. A common but inefficient approach is "One-Variable-At-a-Time" testing, which fails to capture the interactions between different parameters. To overcome this limitation, DoE is employed to systematically sample the parameter space.

The CCD is one of the most widely used frameworks in DoE for constructing a quadratic model of a system without requiring the extensive number of test runs demanded by a full three-level factorial experiment [12]. As illustrated in Figure 4.1, a CCD is comprised of three distinct components:

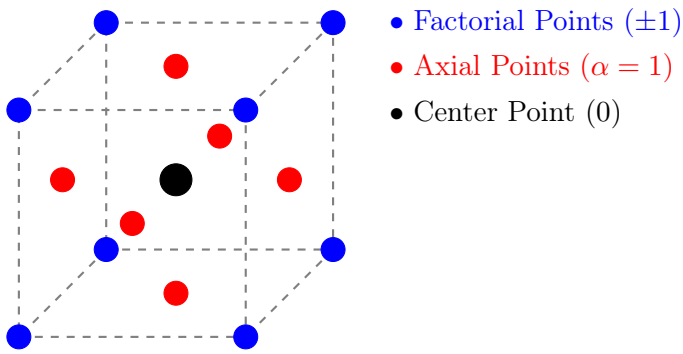

Figure 4.1: Visualization of the three components of a CCD.

1. **Factorial Points:** These represent the corners of a "cube" in the parameter space, testing the combinations of the minimum (-1) and maximum ($+1$) levels for each factor.
2. **Center Points:** These are runs where all factors are set to their median values (0). They are used to determine experimental error and the stability of the system at the nominal setting.

3. **Axial (Star) Points:** These points are located at a distance α from the center. They allow the model to estimate curvature by sampling along the axes, which is essential for identifying optimal peaks or valleys in the response surface.

The total number of experiments for k factors is calculated as follows:

$$N(k) = 2^k + 2k + n \tag{4.3}$$

where N is the total number of experiments, 2^k represents the number of factorial points, $2k$ is the number of axial points, and n is the number of center points [12].

4.3.1 Face-Centered Central Composite Design (FCCD)

In this project, a specific variation known as a Face-Centered CCD is utilized. The defining characteristic of this design is that the axial distance α is set to 1. This configuration places the star points exactly on the centers of the faces of the experimental cube, rather than extending outside of it. While other CCD variations may require five different levels for each parameter, the Face-Centered design only requires three: Minimum (-1), Center (0), and Maximum ($+1$) [12]. This significantly simplifies the implementation of the test bench while still providing sufficient data to map nonlinear interactions between the controller parameters.

By utilizing this FCCD structure, the sensitivities of the MPC controllers can be mapped efficiently, providing a rigorous foundation for comparing their robustness and performance.

4.4 Performance Metrics

To quantitatively evaluate and compare the tracking performance of the MPC controllers, the RMSE is employed. The RMSE provides a measure of the average magnitude of the error between the system state x and the reference x_{ref} over a simulation period of T samples:

$$\text{RMSE} = \sqrt{\frac{1}{T}\sum_{k=1}^{T}(x_k - x_{ref,k})^2} \tag{4.4}$$

This metric is particularly useful for this study as it heavily penalizes large outliers, ensuring that any significant deviations from the height trajectory are reflected in the performance score.

4.5 Experiments

This section outlines a series of experiments used to evaluate the performance of the `acados`-based MPC implementations. Each case study includes a technical description, the underlying rationale for the test, and an interpretation of the results.

4.5.1 Experiment 1: Computational Efficiency Benchmarking

As discussed in Section 4.2, computational time is an important metric for real-time control. This experiment evaluates and compares the real-time feasibility of the MPC controllers using `acados` to both the `CasADi`-based MPC controllers and the RL controllers.

In order to compare the different controllers in an objective way, the same hardware is used in all cases (AMD Ryzen 9 5950X, 128GB RAM, NVIDIA GeForce 710 (2GB)). Furthermore, randomly valid state and target inputs are generated to eliminate eventual bias towards specific trajectories. In the experiment, 1000 such state-target pairs are generated, and the average time it takes to provide an input force is calculated.

Furthermore, the computation time for a MPC controller is heavily dependent on the prediction horizon length N. To investigate how the computation time scales with N, the experiment is repeated for $N \in \{5, 10, 20, 40\}$.

Lastly, the computational efficiency of the `acados`-based MPC controllers are compared against the `CasADi`-based controllers to find the speedup ratio between the implementations. Furthermore, these results are compared to the time step, and thus the real-time feasibility, of both the simulation of the levitating ball and of the psychical Maggy.

4.5.2 Experiment 2: Trajectory Tracking Performance

In this experiment the controllers are subjected to increasingly complex reference trajectories. The purpose of the experiment is to investigate and analyze the performance and limitations of the controllers. Two different types of reference trajectories are considered:

4.5.2.1 Sigmoid Step Response

The sigmoid step response is used to evaluate the controllers ability to transition rapidly between setpoints without inducing overshoot. Unlike a traditional square step response, a sigmoid reference provides a smooth, differentiable trajectory, which is more relevant due to physical limitations such as saturation limits on the input available force. The sigmoid step reference is defined as:

$$r(t) = \frac{C - A}{1 + e^{-k \sin(\omega(t-\phi))}} + 2A \tag{4.5}$$

where C is a scaling factor for the peak height, A is the step magnitude, k is a measures of the steepness of the slope, ω is the step frequency and ϕ is the phase shift. The experiment utilizes the following values for these parameters:

From Equation (4.5), with the parameters from Table 4.2, the sigmoid reference height varies from $2A = 100m$ when the exponential term grows large to $C - A + 2A = 350m$ when the exponential term is zero. Furthermore, given the angular frequency $\omega = 0.3$ rad/s, the period is calculated as $t_f = \frac{2\pi}{\omega} \approx 20.94$ s. Accounting for the phase shift $\phi \approx 5.24$ s, a 30-second window is therefore chosen in order

Table 4.2: Parameters for the Sigmoid Reference Trajectory

Parameter	Symbol	Value	Unit
Peak Height Scaling	C	300	m
Step Magnitude	A	50	m
Steepness Factor	k	$\{5, 10, 15\}$	
Step Frequency	ω	0.3	rad/s
Phase Shift	ϕ	$\frac{\pi}{0.6}$	rad

to capture a complete cycle. This allows for an analysis of the rising and falling edges, which is critical given the asymmetric dynamics caused by the constant gravitational force.

4.5.2.2 Sinusoidal Reference

Sinusoidal trajectories are used to evaluate the controller's tracking performance with dynamic references. A key objective is to estimate the system bandwidth by identifying the frequency limits at which the controller can track the reference without significant phase lag or amplitude attenuation. The sinusoidal reference has the following mathematical formulation:

$$r(t) = A\sin(\omega t + \phi) + D \tag{4.6}$$

where A is the amplitude, ω is the angular frequency, ϕ is the phase shift, and D is the vertical shift. The following values are used:

Table 4.3: Parameters for the Sinusoidal Reference Trajectory

Parameter	Symbol	Value	Unit
Amplitude	A	50	m
Angular Frequency	ω	$\{0.1, 1, 10\}$	rad/s
Phase Shift	ϕ	0	rad
Vertical Shift	D	300	m

The angular frequencies are selected based on the theoretical maximum upward acceleration of the ball. Based on Newton's Second Law, this is defined as $a_{max} = \frac{F_{max}-G}{m}$. Using the nominal parameters from Section 4.1.1, the system's limit is $a_{max} = 40.19$ m/s^2. To perfectly track a sinusoidal reference, the controller must be able to provide a peak acceleration of $A\omega^2$. To assess the controller's performance relative to these physical constraints, the angular frequencies ω listed in Table 4.3 were selected to represent three distinct cases:

1. $\omega = 0.1$ rad/s: A case where the required acceleration ($a_{req} = 0.5$ m/s^2) is negligible compared to a_{max}, allowing for an evaluation of ideal tracking capabilities.

2. $\omega = 1.0$ rad/s: A moderate frequency where the required acceleration ($a_{req} = 50$ m/s^2) slightly exceeds the system's upward limit. This case is chosen to

evaluate how the controller handles the onset of saturation and the resulting trade-off between phase lag and amplitude attenuation.

3. $\omega = 10$ rad/s: A severe stress test where the required acceleration ($a_{req} = 5000$ m/s^2) far exceeds the physical capabilities of the actuator. This scenario is intended to observe the MPC's behavior under extreme infeasibility and its ability to maintain stability while operating at constant saturation.

4.5.2.3 Performance Metric

To evaluate the deviation from the reference, RMSE is used as the performance metric as described in Section 4.4. Since both slow convergence, latency and overshoot lead to a deviation between the actual height and the desired trajectory, all these effects are captured by the RMSE. However, a more complex performance metric could have been used to penalize the effects differently, e.g. to penalize overshoot more severely.

4.5.3 Experiment 3: Parameter Sensitivity and Optimization via Response Surface Methodology (RSM)

This experiment seeks to replace heuristic "trial-and-error" tuning with a rigorous statistical framework using RSM. The objective is to map the controller's performance landscape and identify the optimal configuration for the tuning parameters.

4.5.3.1 Design Objectives

The experimental design is structured to address three primary questions:

- **Horizon Impact:** How does the prediction horizon length (N) influence the trade-off between tracking precision and computational cost?
- **Weighting Sensitivity:** How do variations in the state ($\mathbf{Q}$) and control ($\mathbf{R}$) weighting matrices affect stability?
- **Parametric Robustness:** How sensitive is the closed-loop performance to small deviations in these tuning parameters?

4.5.3.2 Methodology and DoE Framework

The tuning process begins with a baseline parameter set derived from Marius, followed by a coarse heuristic refinement to establish a viable search space. Within this space, a Face-Centered CCD is employed as described in Section 4.3, with the following tuning parameters:

The rationale behind the selection of the first 4 parameters is described in Section 4.5.3.1 as $\mathbf{Q} = diag(q_h, qv)$ and $\mathbf{R} = r$. Ultimately, $r_{\Delta u}$ is included to penalize rapid changes in the control input, as described in Section 3.2.

Applying Equation (4.3), a CCD with five parameters and a single center point results in 43 samples. While such a high sample count is often too time-consuming for physical experiments, the computational efficiency of `acados` allows for rapid

Table 4.4: Tuning Parameters for the CCD Optimization.

Parameter	Symbol	Description
Horizon Length	N	Look-ahead distance
Height Weight	q_h	Penalty on position error
Velocity Weight	q_v	Penalty on ball speed
Input Weight	r	Penalty on absolute force
Slew Rate Weight	$r_{\Delta u}$	Penalty on change in force

simulation. This enables a high-resolution mapping of the input-output relationship within a reasonable time frame.

Unlike the traditional One Variable At a Time (OVAT) approach, which varies one parameter while holding others constant, the CCD captures the interaction effects between variables. This is particularly relevant for MPC, where the ratio of weights often dictates behavior more significantly than their absolute magnitudes. The DoE approach also enables the identification of the parameters with the most significant impact on system output.

4.5.3.3 Test Trajectories and Optimization Metric

To ensure the resulting parameters are robust, the controller is evaluated against two sigmoid reference signals of varying difficulty ($k \in \{10, 15\}$). The optimization objective is the minimization of the average RMSE across both trajectories. This dual-signal approach prevents "over-fitting" the controller to a single trajectory and ensures a balanced performance across the system's dynamic range. The final output of this design will be presented as contour plots to visualize both the optimal settings and the robustness to parameter perturbations.

4.5.4 Experiment 4: Robustness Analysis under Model Mismatch

This experiment introduces a deviation between the nominal mass (m_{nom}) assumed by the MPC and the true physical mass of the ball ($m_{\text{ball}} = 1.0$ kg). The objective is to evaluate the controller's robustness and determine its tolerance to significant model mismatches. Two scenarios are investigated, where $m_{\text{nom}} = m_{\text{ball}} \pm 0.7 m_{\text{ball}}$:

1. **Underestimation** ($m_{\text{nom}} = 0.3$ kg): This case is expected to result in a steady-state offset, as the controller provides insufficient force to counteract gravity.
2. **Overestimation** ($m_{\text{nom}} = 1.7$ kg): This is tested to observe potential overshoot and oscillations, as the controller may overreact to tracking errors by commanding excessive restorative force.

CHAPTER

FIVE

RESULTS

This chapter presents the experimental results for the four experiments defined in Section 4.5. A comprehensive analysis and interpretation of these findings are provided in Section 6.1.

5.1 Experiment 1: Computational Efficiency Benchmarking

This experiment evaluates and compares the real-time feasibility of the MPC controllers, as described in Section 4.5.1. Figure 5.1 presents the average computational time for the three different `acados`-based controllers. All controllers show a near-linear relationship between the prediction horizon N and the average computational time. Table 5.1 shows the computational efficiency comparison between the `acados`-based and `CasADi`-based controllers, highlighted by the ratio column. The ratios indicate that the `acados`-based implementation is significantly faster across all test cases, consistently outperforming the `CasADi` alternative by a substantial margin.

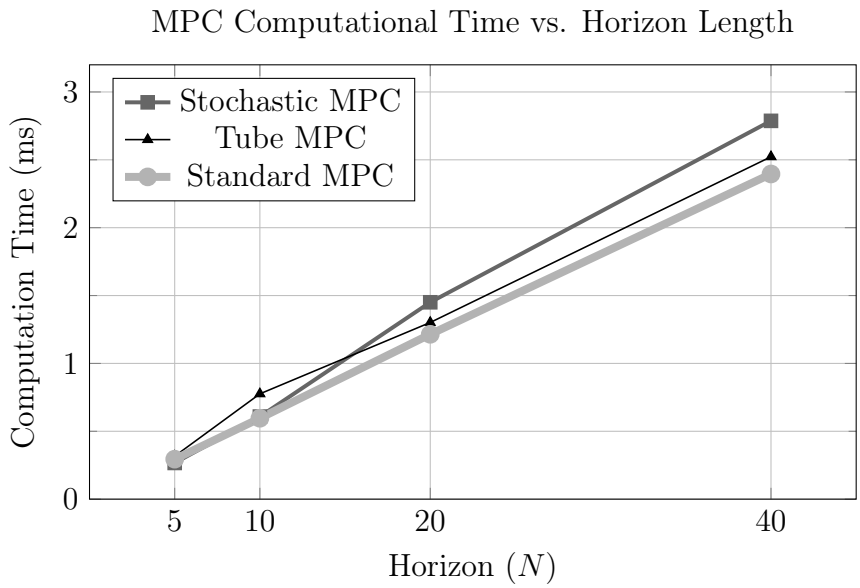

Figure 5.1: Computational performance comparison for varying horizons N.

Controller	**Horizon** (N)	`acados` **(ms)**	`CasADi` **(ms)**	**Ratio** ($\times$)
Standard MPC	5	0.2945	6.5581	22.3
	10	0.5961	6.6611	11.2
	20	1.2146	10.1335	8.3
	40	2.3957	18.2578	7.6
Stochastic MPC	5	0.2666	9.3707	35.1
	10	0.6085	14.1025	23.2
	20	1.4504	26.9964	18.6
	40	2.7871	61.1214	21.9
Tube MPC	5	0.3165	4.6344	14.6
	10	0.7758	6.4479	8.3
	20	1.3016	8.8715	6.8
	40	2.5226	16.7322	6.6

Table 5.1: Computational Time Comparison: `acados` vs. `CasADi` (ms)

5.2 Experiment 2: Trajectory Tracking Performance

This experiment evaluates the trajectory tracking performance of the MPC controllers on a series of reference trajectories of varying complexity.

Figure 5.2 illustrates the MPC tracking performance for the sigmoid trajectories defined in Section 4.5.2.1, with the corresponding output force displayed on the right side of the figure. The figure shows that an increase in the slope k yields a higher degree of overshoot and longer settling time for all 3 controllers. This is captured by the resulting RMSE, which is presented in Table 5.2.

Similarly, Figure 5.3 illustrates the tracking performance and control effort for the sinusoidal references defined in Section 4.5.2.2. At the lowest angular frequency, the controllers exhibit near-perfect tracking. Conversely, the high-frequency case reveals the system's bandwidth limitations, where the controllers are unable to follow the rapid oscillations due to saturation limits of the available force. The corresponding RMSE values are listed in Table 5.3.

Figure 5.4 shows the comparison between the `acados`-based standard MPC, the `CasADi`-based standard MPC, and the RL performance on a sigmoid trajectory with $k = 10$. As seen in Table 5.4, the `acados`-based MPC yields the lowest RMSE among the controllers.

Slope	**Standard MPC**	**Stochastic MPC**	**Tube MPC**
5	37.35110	26.98699	28.97125
10	59.29391	44.91656	56.09871
15	73.43328	62.79734	72.47844

Table 5.2: RMSE for various MPC-controllers for sigmoid trajectories.

Angular Frequency	**Standard MPC**	**Stochastic MPC**	**Tube MPC**
0.1 rad/s	49.79478	49.22609	53.09393
1 rad/s	65.10046	63.76868	68.59049
10 rad/s	61.38448	62.58449	66.27469

Table 5.3: RMSE for various MPC-controllers for sinusoidal trajectories.

Controller	**RMSE**
`Acados` MPC	59.29391
`CasADi` MPC	65.40366
RL	66.09262

Table 5.4: RMSE comparison of `acados` MPC, `CasADi` MPC, and RL on sigmoid trajectory with slope 10.

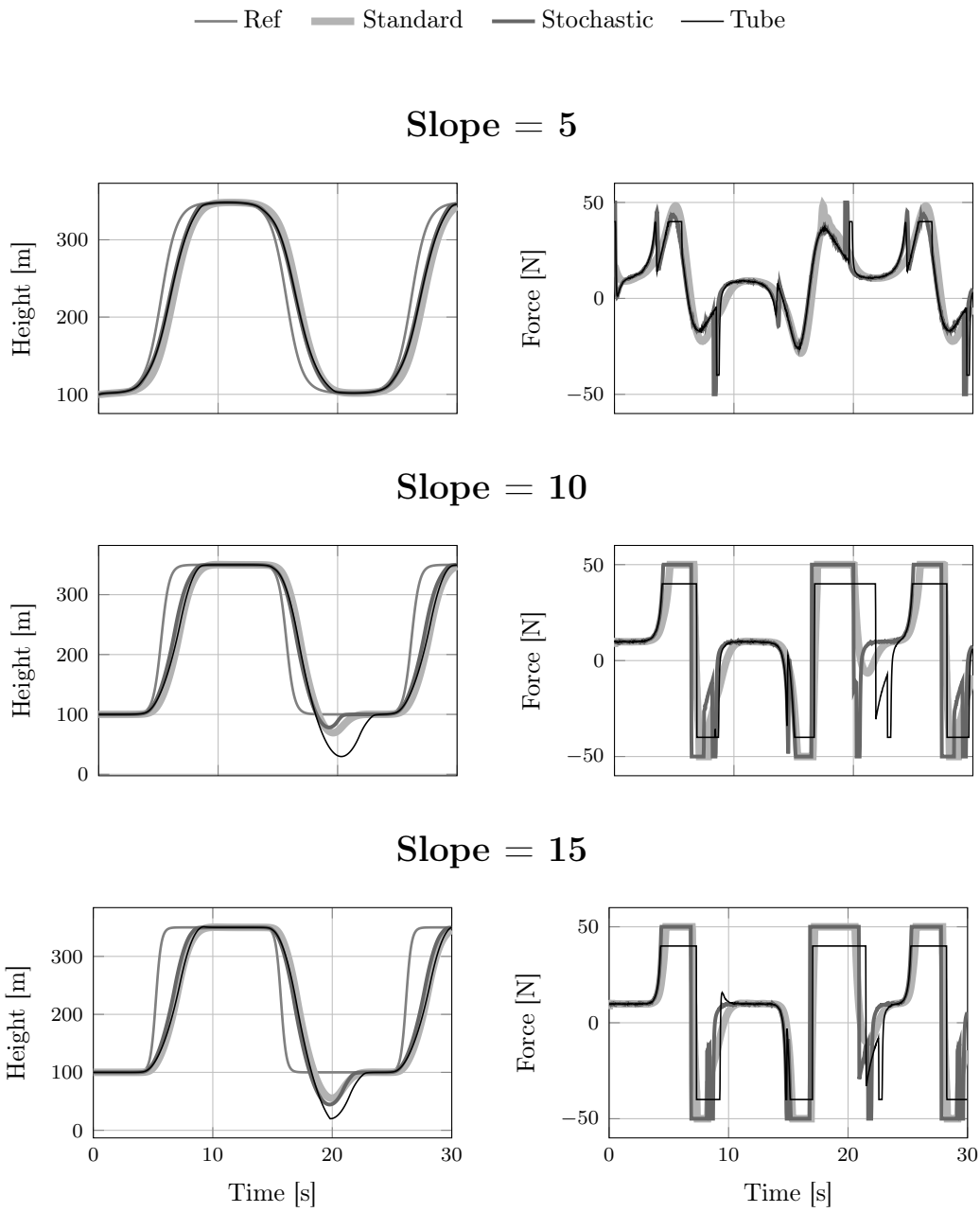

Figure 5.2: Comparison of MPC-controllers for sigmoid trajectories with slopes 5, 10, and 15.

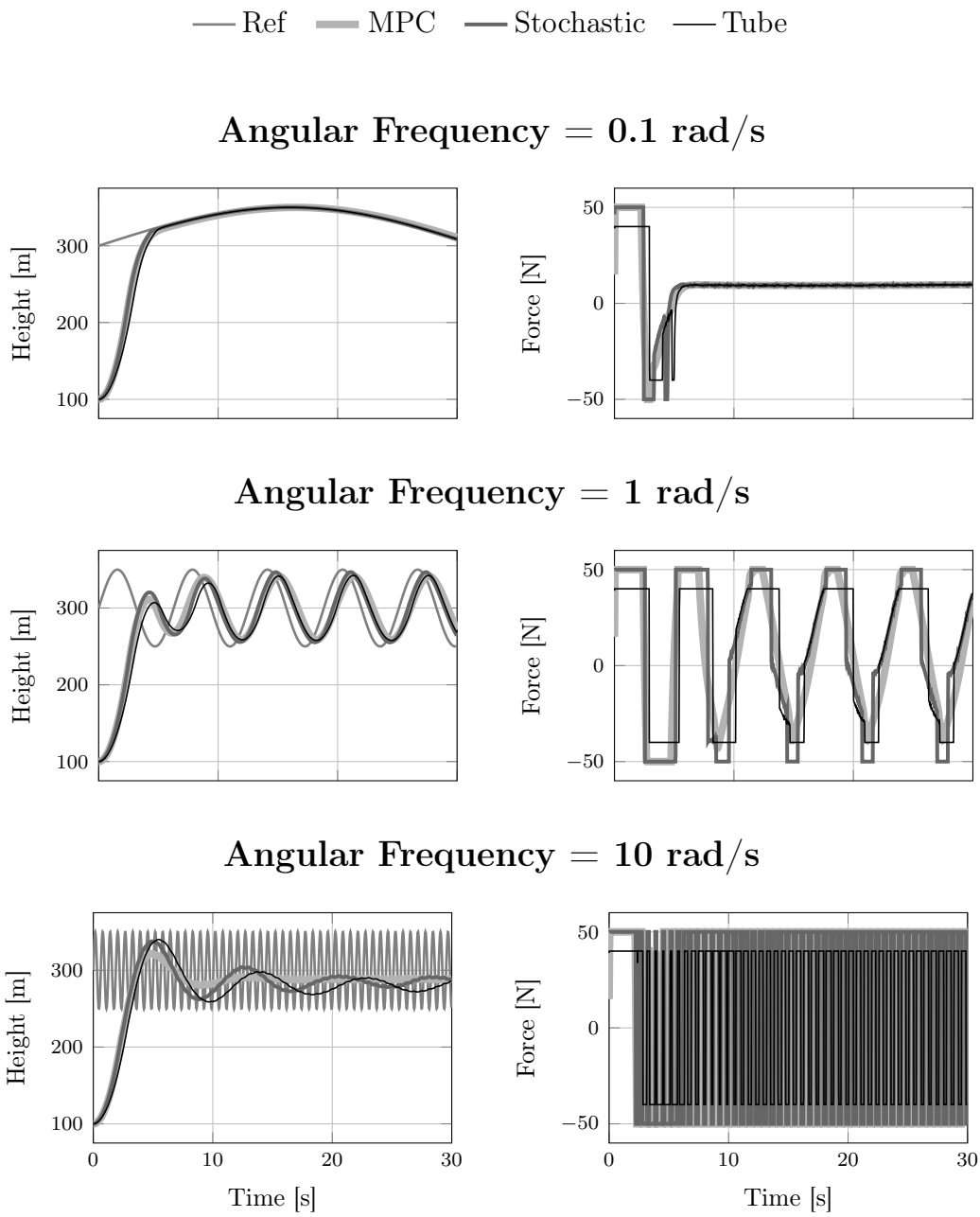

Figure 5.3: Comparison of MPC-controllers for sinusoidal trajectories with angular frequencies 0.1, 1, and 10.

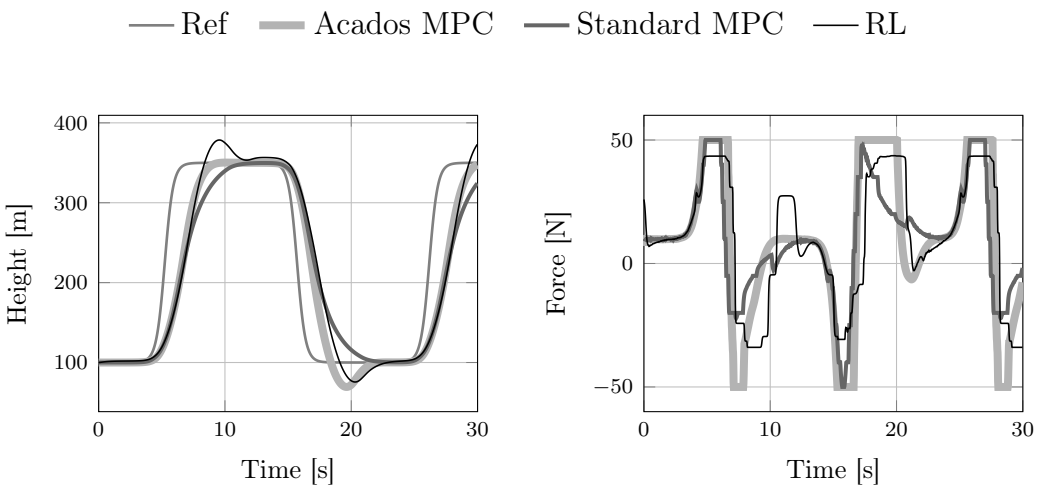

Figure 5.4: Comparison of `acados` MPC, `CasADi` MPC, and RL performance on sigmoid trajectory with slope 10.

5.3 Experiment 3: Parameter Sensitivity and Optimization via Response Surface Methodology (RSM)

As described in Section 4.5.3, the initial tuning was based on a parameter set derived for the Marius's `CasADi`-based MPC, which is listed in Table 5.5. When using these parameters, the `acados` controllers exhibited excessive overshoot and high-frequency oscillations (chatter) in the force commands. To reduce the aggressiveness of the controller, the weight q_h was reduced from 100 to 70, balancing the requirement for short settling times with the need to minimize overshoot. Furthermore, to ensure smoother control effort, a penalty on the rate of change of the input, $r_{\Delta u}$, was incorporated into the cost function. Through iterative tuning, $r_{\Delta u}$ was increased until a value of 50 was reached, at which point the controller provided consistently smooth actuator commands across all tested trajectories.

Parameter	**Value**
Horizon (N)	20
q_h	100
q_v	20
r	0.05

Table 5.5: `CasADi` MPC parameters.

As described in Section 4.5.3, a CCD is then used to find the optimal configuration of tuning parameters and investigate the parametric robustness. The minimum values in the CCD are set to 0.5 times the center values to balance the trade-off between searching the parameter space and detecting curvature. Table 5.6 shows the resulting CCD, where the maximum values are 1.5 times the center values to

ensure symmetry.

Parameter	Minimum (-1)	Center Value (0)	Maximum (+1)
N	10.0	20.0	30.0
q_h	35.0	70.0	105.0
q_v	10.0	20.0	30.0
r	0.025	0.05	0.075
$r_{\Delta u}$	25.0	50.0	75.0

Table 5.6: Central Composite Design (CCD) levels for MPC Tuning.

The CCD is used on the sigmoid trajectories defined in Section 4.5.2.1 with $k \in \{10, 15\}$ to calculate the RMSE for the 43 different cases. The outputted RMSE is then used to generate a quadratic model of the response surface.

The resulting Analysis of Variance (ANOVA) table for $k = 10$ is presented in Table 5.7. The table shows that the model fits the data well as it accounts for more than 99% of the explained variance. It furthermore shows that changes in the state weights q_h and q_v are the main factors for variations in the outputted RMSE, while the factors $r_{\Delta u}, N,$ and r are negligible.

Source of Variation	Sum of Squares	Contribution (%)	Significance
q_h	1275.08	42.43%	High
q_v	1113.72	37.06%	High
$q_h \times q_v$	487.92	16.24%	High
q_h^2	50.34	1.68%	Moderate
q_v^2	0.08	0.00%	Negligible
$r_{\Delta u}, N, r$	3.26	0.11%	Negligible
Residual (Error)	17.76	0.59%	—
Total	3004.81	100.00%	

Table 5.7: ANOVA breakdown of factor importance for MPC RMSE for slope 10.

Figure 5.5 shows the contour plot for $k = 10$, with state weights q_h and q_v against the RMSE. The figure indicates that the optimal setting for this specific trajectory is outside of the search space, with a combination of high q_h and low q_v.

Table 5.8 shows the ANOVA table for $k = 15$. The table shows that the quadratic model fits accounts almost 99% of the explained variance. In contrast to the previous case, the interaction term $q_h \times q_v$ is now the main factor for the variations in the outputted RMSE. Similar to the previous case, $r_{\Delta u}, N,$ and r remains negligible.

Figure 5.6 shows the contour plot for $k = 15$, with q_h and q_v against RMSE. The figure indicates that the optimal setting for the sigmoid trajectory is much more conservative than for $k = 10$, as the combination of high q_h and low q_v results in high RMSE values.

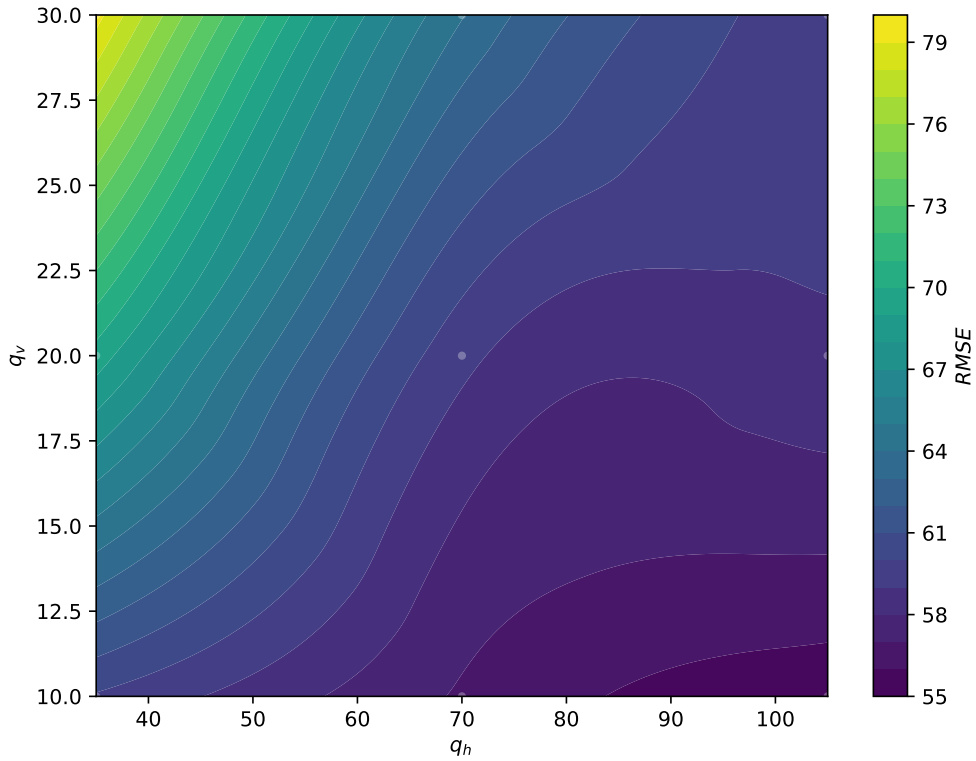

Figure 5.5: Contour plot showing the state weights q_h and q_v against RMSE for slope 10

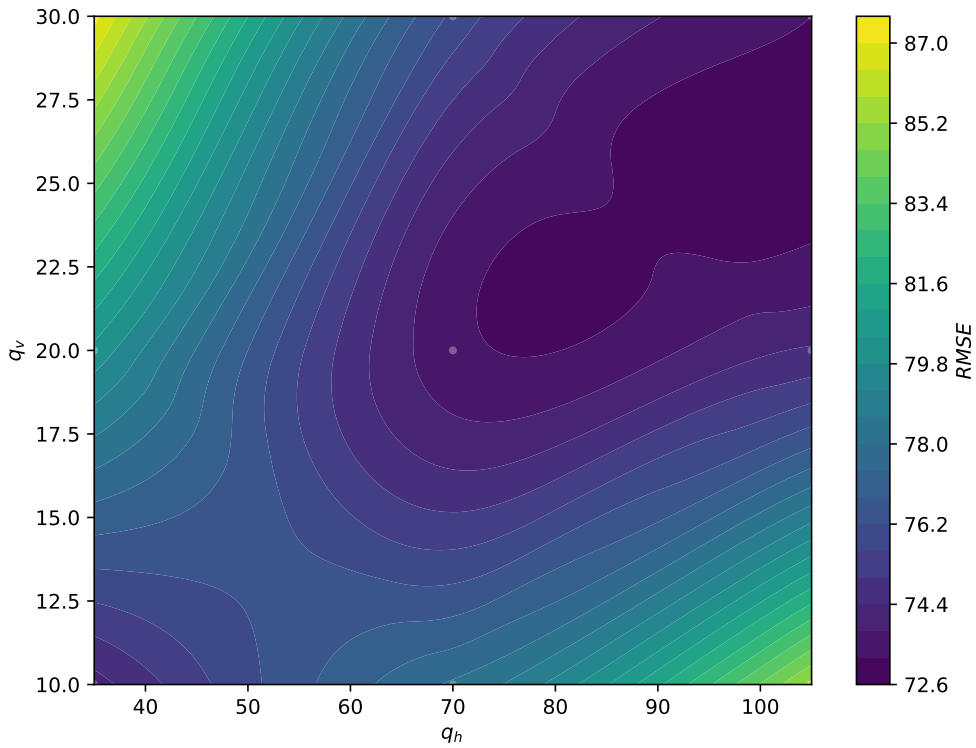

Figure 5.6: Contour plot showing the state weights q_h and q_v against RMSE for slope 15

Source of Variation	Sum of Squares	Contribution (%)	Significance
q_h	43.79	3.01%	Moderate
q_v	32.17	2.21%	Moderate
$q_h \times q_v$	1279.68	87.95%	High
q_h^2	39.16	2.69%	Moderate
q_v^2	33.16	2.28%	Moderate
$r_{\Delta u}, N, r$	11.05	0.76%	Negligible
Residual (Error)	15.95	1.10%	—
Total	1454.95	100.00%	

Table 5.8: ANOVA breakdown of factor importance for MPC RMSE for slope 15.

To make the controller robust to different trajectories, the average RMSE of the two sigmoid trajectories is used as the performance metric. This yields the contour plot seen in Figure 5.7. The figure shows that a high q_h and a low to moderate q_v consistently provides the lowest error.

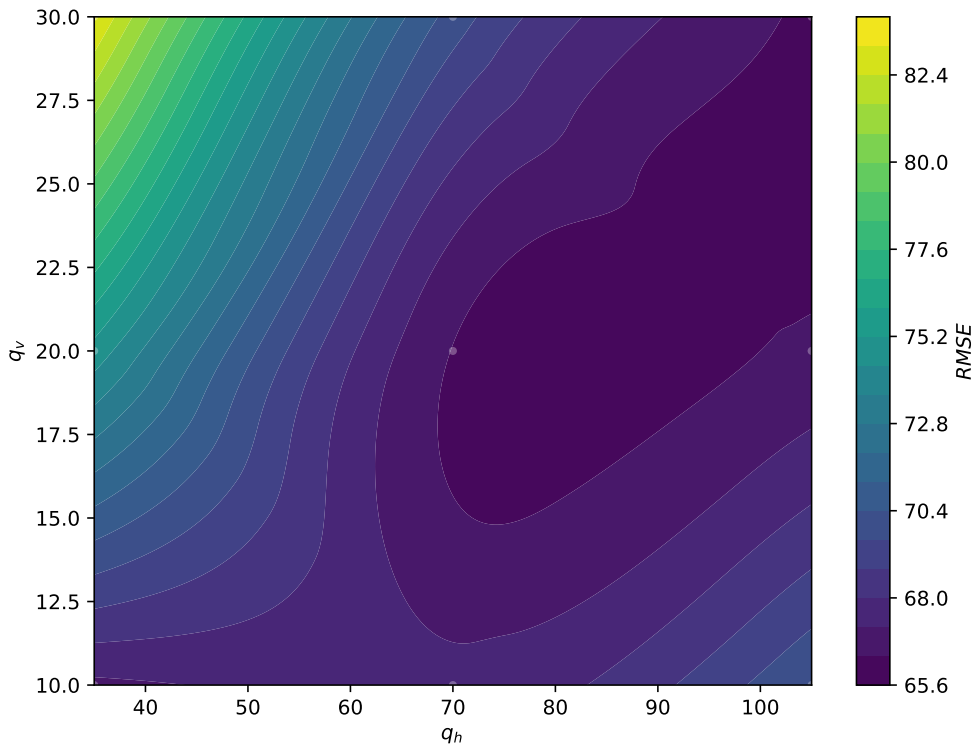

Figure 5.7: Contour plot showing the state weights q_h and q_v against the average RMSE

5.4 Experiment 4: Robustness Analysis under Model Mismatch

This experiment evaluates the robustness of the MPC controllers by introducing model mismatch, as outlined in Section 4.5.4.

The results from the underestimation case, where the assumed mass is 0.3 kg, are shown in Figure 5.8 and Figure 5.9, with the corresponding RMSE values listed in Table 5.9 and Table 5.10. Figure 5.8 shows that all controllers have problems stabilizing on the flat parts of the sigmoid step trajectories, especially for higher slopes. Compared to Figure 5.3, where the controllers assume the correct mass, Figure 5.9 shows that the underestimation of the mass introduces additional phase lag for the sinusoidal trajectory with angular frequency $\omega = 1.0$ rad/s.

The results from the overestimation case, where the assumed mass is 1.7 kg, are similarly shown in Figure 5.10 and Figure 5.11, with the corresponding RMSE values listed in Table 5.11 and Table 5.12. Unlike the underestimation case, the performance is very similar to the case with perfect knowledge of the model, as seen in Figure 5.2 and Figure 5.3. This close alignment with the ideal model case demonstrates that the controllers remain robust under these conditions.

Slope	**Standard MPC**	**Stochastic MPC**	**Tube MPC**
5	44.42780	13.58351	18.13449
10	69.13154	48.77723	55.83441
15	81.06954	65.28128	72.62342

Table 5.9: RMSE with lighter ball for sigmoid trajectories.

Angular Frequency	**Standard MPC**	**Stochastic MPC**	**Tube MPC**
0.1 rad/s	49.71419	49.47459	53.18785
1 rad/s	78.60607	92.78596	89.79911
15 rad/s	61.64231	66.23357	68.91873

Table 5.10: RMSE with lighter ball for sinusoidal trajectories.

Slope	**Standard MPC**	**Stochastic MPC**	**Tube MPC**
5	37.04937	30.73019	32.23644
10	57.84656	48.53261	57.70923
15	72.30148	62.94376	73.32282

Table 5.11: RMSE with heavier ball for sigmoid trajectories.

Angular Frequency	**Standard MPC**	**Stochastic MPC**	**Tube MPC**
0.1 rad/s	49.94832	49.34616	53.16479
1 rad/s	64.11607	62.75674	66.95129
10 rad/s	61.48381	61.97466	65.57440

Table 5.12: RMSE with heavier ball for sinusoidal trajectories.

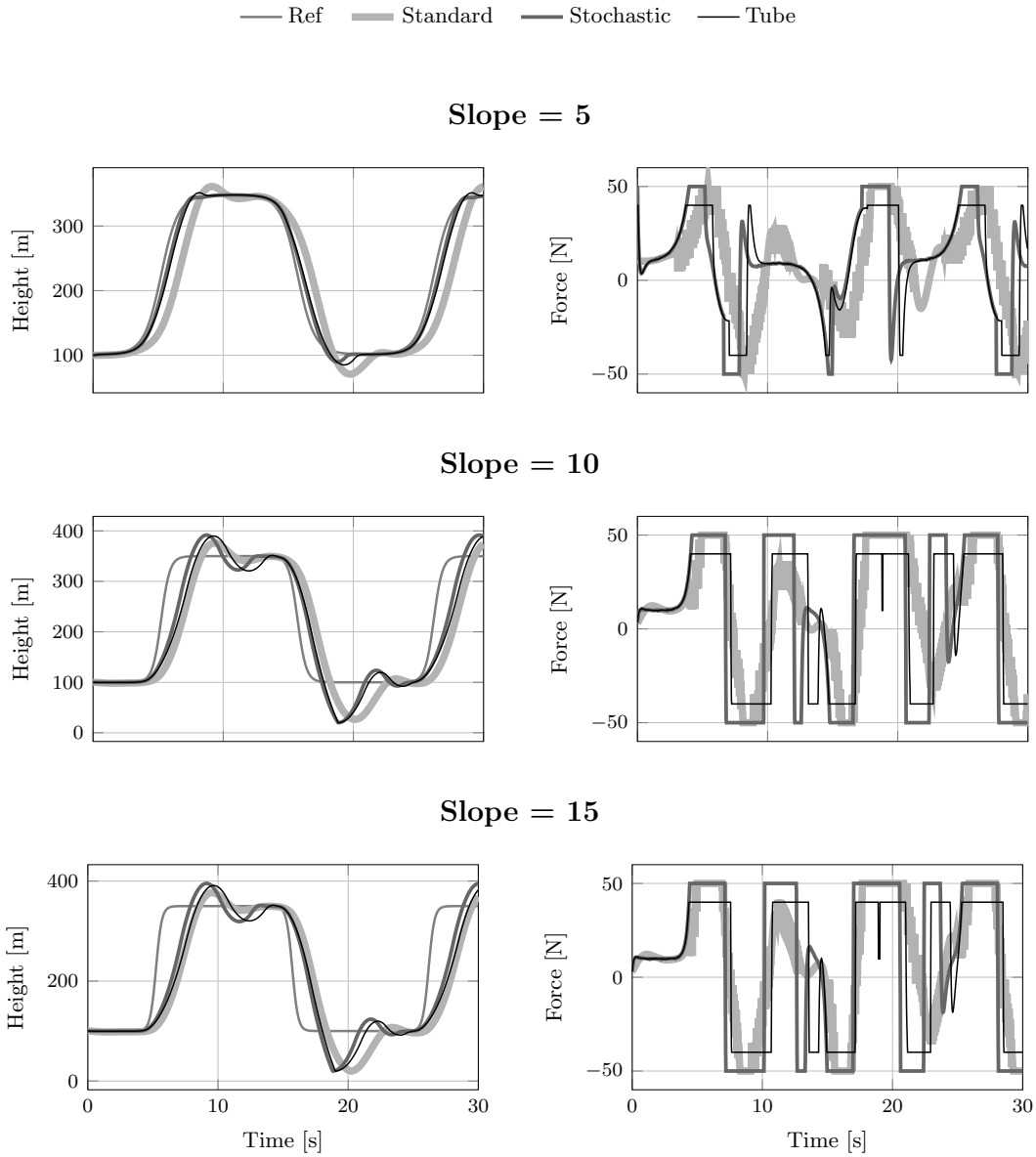

Figure 5.8: Model mismatch with lighter ball for sigmoid trajectories with slopes 5, 10, and 15.

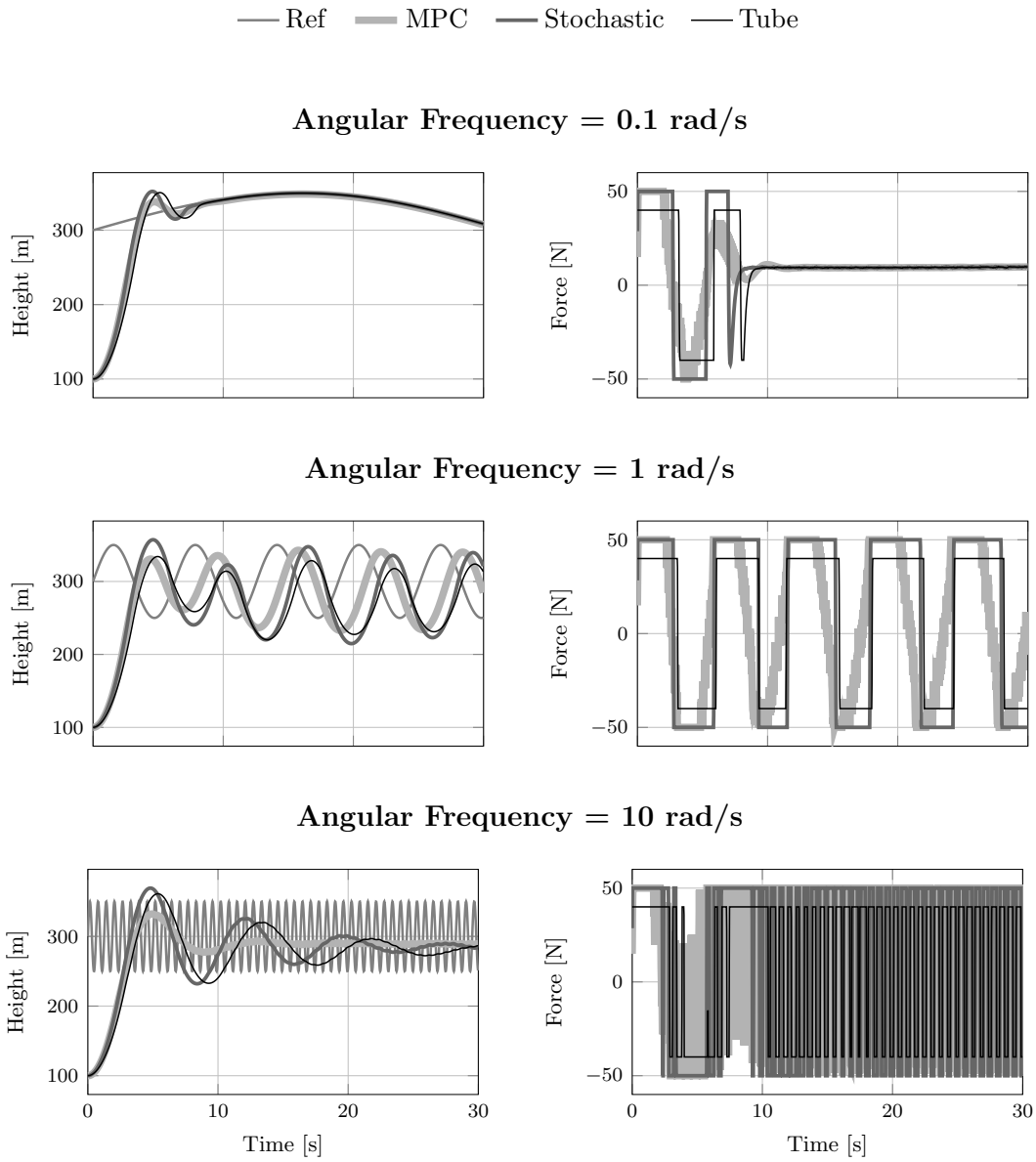

Figure 5.9: Model mismatch with lighter ball for sinusoidal trajectories with angular frequencies 0.1, 1, and 10.

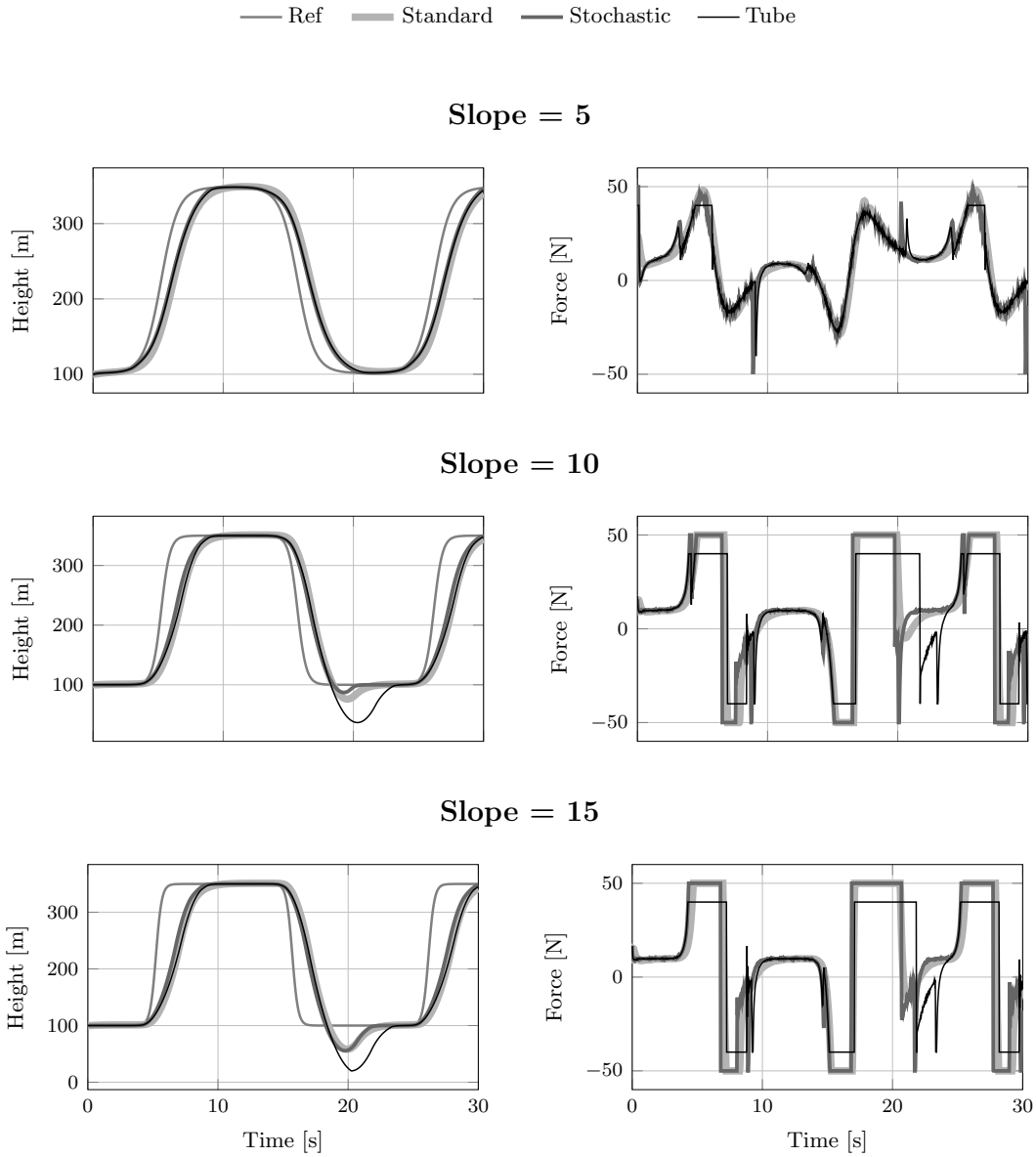

Figure 5.10: Model mismatch with heavier ball for sigmoid trajectories with slopes 5, 10, and 15.

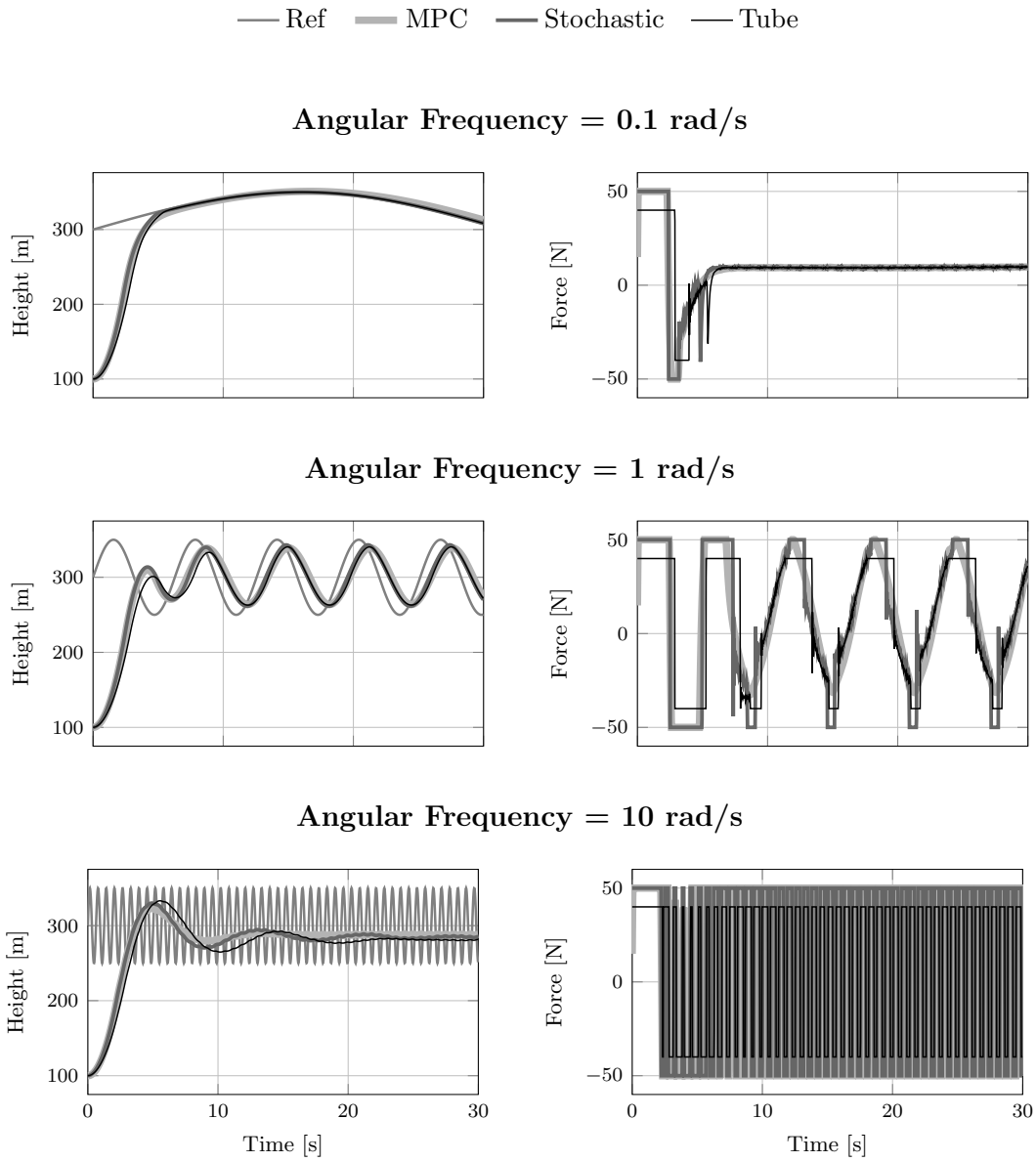

Figure 5.11: Model mismatch with heavier ball for sinusoidal trajectories with angular frequencies 0.1, 1, and 10.

CHAPTER

SIX

DISCUSSION

6.1 Interpretation of results

This section provides a comprehensive analysis of the results presented in Chapter 5. Each subsection in this chapter contains the analysis of a single experiment.

6.1.1 Experiment 1: Computational Efficiency Benchmarking

The benchmarking results presented in Table 5.1 show a large contrast in both execution time and scaling behavior between the `acados`-based and `CasADi`-based implementations.

6.1.1.1 Prediction Horizon N Scaling

A primary objective of this experiment was to investigate how the computational efficiency scales with the prediction horizon N. As illustrated in Figure 5.1, all `acados` controllers exhibit a linear scaling relationship ($O(N)$). For the standard `acados` MPC, doubling the horizon from $N = 20$ to $N = 40$ results in a near-perfect doubling of execution time, from 1.2146 ms to 2.3957 ms. As discussed in Section 4.2, the linearity of the solvers utilized by `acados`, specifically `HPIPM`, is partially caused by the exploitation the block-sparse structure of the OCP.

In contrast, the `CasADi` implementations exhibit less efficient scaling, particularly in the stochastic case. For the stochastic `CasADi` MPC, increasing N from 20 to 40 leads to an increase in computation time from 26.9964 ms to 61.1214 ms. This indicates a superlinear scaling penalty, likely due to the overhead of the general-purpose `IPOPT` solver when handling the increased dimensionality of the symbolic expressions and constraints as N grows.

6.1.1.2 Real-Time Feasibility

The feasibility of the controller is determined by whether the computation time is strictly less than the sampling time Δt. Furthermore, reducing the computational time further is vital, as higher sampling frequencies improve disturbance rejection and overall system stability.

- **Levitating Ball ($\Delta t = 20$ ms):** All `acados` and most `CasADi` controllers satisfy this constraint. Only the stochastic `CasADi` MPC at $N = 20$ (27.00 ms) and $N = 40$ (61.12 ms) fails to meet the real-time requirements for the 1-DOF system.
- **Physical Maggy System ($\Delta t \approx 3$ ms):** The sampling time for the Maggy system is based on the sampling time used in the simulations in Mikael's master thesis [3]. This imposes significantly more stringent demand on the computational efficiency. The `CasADi` implementation fails this requirement across all test cases, with even the simplest horizon ($N = 5$) requiring 6.5581 ms. This is without considering that the computational cost would increase for the Maggy system compared to the 1-DOF system, due to the increased dimensionality and nonlinear dynamics. Conversely, all `acados` variants remain below the 3 ms threshold for all tested horizons, which indicates greater potential for real-time feasibility.

6.1.1.3 Comparison of The `acados` MPC Controllers

Figure 5.1 shows that the inclusion of robustness (Tube MPC) or chance constraints (Stochastic MPC) in `acados` does not significantly increase the computation time. While the Tube MPC at $N = 10$ (0.7758 ms) is slightly slower than the Standard MPC (0.5961 ms), the difference is small in the context of real-time control. This suggests that the trade-off between control complexity and computational cost can be well balanced, allowing for advanced robust formulations without sacrificing the high sampling rates required to stabilize the Maggy system.

6.1.2 Experiment 2: Trajectory Tracking Performance

The results from this experiment show that the performance of the `acados` MPC controllers is comparable to the `CasADi` MPC implemented by Marius. Furthermore, the results provide insights into the trade-off between tracking performance and the control effort.

6.1.2.1 Comparison of Control Strategies

Figure 5.4 shows the performance of three different control strategies: `acados` MPC, `CasADi` MPC and RL. During the initial ascent of the sigmoid slope, the `acados` MPC is superior as it exhibits a shorter settling time than the `CasADi` MPC. Furthermore, the RL controller induces a slight overshoot before converging to the steady-state height. During the descent, all controllers achieve similar convergence times. However, `acados` and RL both suffer from noticeable overshoot, whereas the `CasADi` MPC provides a superior response by reaching the target height smoothly. In total, as seen in Table 5.4, the `acados` MPC yields the lowest tracking error

with a RMSE of 59.29, compared to 65.40 (`CasADi MPC`) and 66.09 (RL). If the performance metric had been adjusted, by also penalizing absolute control effort, changes in control effort or penalizing overshoot more severely, the `CasADi` would likely have been yielded the best performance. However, the differences seen by the `acados` and `CasADi` controllers are likely primarily seen due to tuning and not the inherent solvers. This conclusion is based on the heuristic tuning done, which showed a similar trade-off between settling time and the amount of overshoot.

6.1.2.2 `acados` MPC Controller Performance

Figure 5.2 and Figure 5.3 shows the performance of the three `acados`-based controllers on the sigmoid and sinusoidal trajectories, respectively. As expected, an increase of the slope of the sigmoid step trajectory, resulted in increasingly more error as seen by the RMSE values listed in Table 5.2. Noticeably, the controllers yield no visibly overshoot in the ascent, but increasingly more overshoot with steeper slopes in the descent. This is a result of the non-symmetric dynamics, where the maximal absolute force downwards is $F_{\max} + G = 59.81$ N and $F_{\max} - G = 40.19$ N upwards. The RMSE values suggest that the Standard MPC performs the worse. However, as mentioned in Section 1.6, these controllers cannot be directly compared to the Standard MPC, as only the Standard MPC constrains the maximum change in force between two time steps. This can be visualized in the outputted force for the Stochastic and Tube MPC for slope 10 and 15, where the force changes instantaneous from minimum to maximum applicable value in a single time step. For the sinusoidal wave with angular frequency 0.1, all controllers are able to converge to and then track the trajectory. The sinusoidal wave with frequency 1 provides a greater challenge as the acceleration of the steepest points exceeds the maximum acceleration the ball, as outlined in Section 4.5.2.2. All controllers yield some phase lag, but little to none amplitude attenuation is observed. Counterintuitively, the RMSE for angular frequency 10 is lower for all controllers compared to angular frequency 1, as seen in Table 5.3. This highlights a weakness with the experimental setup, where the reduced average error in the first 5 seconds of the high-frequency trajectory heavily impacts the total RMSE. This could have been avoided by reducing the initial vertical offset between the ball and reference height.

6.1.3 Experiment 3: Parameter Sensitivity and Optimization via RSM

The application of a Central Composite Design (CCD) allowed for a systematic exploration of the MPC parameter space, revealing that the controller's performance is dominated by a non-linear interaction between the height and velocity weighting. The following subsections discuss the sensitivity to parameter perturbations and the resulting optimal configuration.

6.1.3.1 Influence of Prediction Horizon (N)

As shown in both Table 5.7 and Table 5.8, the prediction horizon length N contributed less than 1% to the total variance of the RMSE. This suggests that within the tested range of $N \in [10, 30]$, the controller had already reached a performance plateau. This suggests that a horizon of 10 or less is sufficient for

this simple system. Repeating the experiment with a lower center value for the horizon could be used to find an optimal value of N which balances computational efficiency and performance.

6.1.3.2 Weighting Matrices and Stability

The sensitivity analysis highlights a sharp contrast between the influence of state weights (q_h, q_v) and the control weights $(r, r_{\Delta u})$.

- **State Weights (Q):** The tracking performance is highly sensitive to q_h (height deviation) and q_v (vertical velocity). In the slope 10 case, linear effects dominated. However, as the trajectory difficulty increased in the slope 15 case, the interaction term $q_h \times q_v$ became the primary source of performance, explaining 87.95% of the variance. This indicates that at steeper slopes, stability is not a function of individual weights, but rather the ratio between the aggressiveness of the height correction (q_h) and the damping provided by the velocity weight (q_v).
- **Control Weights (r and $r_{\Delta u}$):** The input weight r was found to be negligible, suggesting that the tested range of r was chosen too small. Similarly, $r_{\Delta u}$, which represents the penalization of the change in control, exhibited a very low contribution to the total RMSE variance in both cases. However, this lack of influence on the tracking error is a significant finding for controller refinement. It indicates that $r_{\Delta u}$ can be leveraged as a independent tuning parameter to improve the smoothness of the force outputs and reduce mechanical tear. Because the RMSE is insensitive to this parameter within the tested range, $r_{\Delta u}$ can be increased to suppress rapid control fluctuations and ensure a more stable, continuous control force without sacrificing the tracking performance of the levitating ball.

6.1.3.3 Sensitivity to Parameter Deviations

The resulting response surface, visualized in the contour plots in Figure 5.5 and Figure 5.6, reveals the controller's sensitivity to tuning inaccuracies. The valleys in the contour plots represent a robust region where small deviations in tuning parameters do not significantly degrade the RMSE. Conversely, the steep gradients indicate that moving toward lower q_h values causes a rapid decline in performance. The strong interaction observed in the steeper trajectory implies that if one parameter is deviated from its optimal setting, the other must be adjusted proportionally to prevent a sharp increase in tracking error.

6.1.3.4 Optimal Parameter Configuration

The objective of the optimization was to find a robust set of tuning parameters that minimizes RMSE across varying trajectory slopes. Figure 5.7 can be utilized for this selection by averaging the performance across the two sigmoid trajectories.

The optimal configuration is located at the center of the lowest average RMSE valley:

- $q_h \approx 80$ **to** 90**:** High horizontal weighting ensures the ball reaches the target height quickly
- $q_v \approx 20.0$ **to** 25.0**:** Moderate vertical velocity weighting provides the necessary damping to prevent the ball from inducing significant overshoot and oscillations.

This configuration represents the most effective trade-off, ensuring the controller maintains the stability required to track complex dynamic trajectories while still providing fast convergence.

6.1.4 Experiment 4: Robustness Analysis under Model Mismatch

The model mismatch experiment provided valuable insights into the controller's robustness and limitations. The following subsections discuss the two cases of underestimation and overestimation, before concluding based on the overall result.

6.1.4.1 Underestimation Case ($m_{\mathbf{nom}} = 0.3$ **kg**)

Figure 5.8 shows the performance on the sigmoid step trajectories, with the corresponding RMSE values shown in Table 5.9. For the Standard MPC controller, the trajectory with slope 5 introduces overshoot and shows the difficulties to stabilize around the steady-state phases of the trajectory. This is expected, as the assumed force to counteract gravity is significantly less in the underestimation case. For comparison, the real hover force needed to maintain height is 9.81 N, while the assumed hover force is 2.94 N. Moreover, slope 10 induces more severe overshoot for the Standard MPC, especially in the downward phase of the trajectory. However, as the slope increases to 15, the shape of the trajectory is very similar to slope 10. Due to the induced overshoot and oscillations, the Standard MPC yields higher RMSE than in the nominal case (Table 5.2). Noticeably, the RMSE for both the Tube and Stochastic MPC is significantly lower for slope 5 than in the nominal case. However, as the slope increases, the levitating ball crashes into the ground, which is seen by the discontinuity in the height velocity at the bottommost point. Figure 5.9 shows the performance on the sinusoidal trajectories. For angular frequency 0.1, all controllers induce some damped oscillations around the reference before converging. As the angular frequency increases to 1, all controllers introduce additional phase lag compared to the nominal case (Figure 5.3). This is captured by the increased RMSE values, which increases by more than 20% for all controllers.

6.1.4.2 Overestimation Case ($m_{\mathbf{nom}} = 1.7$ **kg**)

Figure 5.10 and Figure 5.11 shows the performance on the sigmoid and sinusoidal trajectories, respectively. In contrast to the underestimation case, the results are quite similar to the nominal case, both in terms of RMSE values and tracking trajectories. Interestingly, for the Standard MPC on the sigmoid trajectory with slope 10, the degree of overshoot is reduced in the overestimation case, and the RMSE is slightly decreased from 59.29 to 57.85. The reduction in overshoot is logical as the controller overestimates the force needed to reduce the velocity of the ball, thus initializing the de-acceleration earlier on.

6.1.4.3 Discussion about Robustness and Tuning

Based on the results, the controllers are more robust to overestimation of the force compared to underestimation. However, the underestimation case deviates more significantly from the nominal response. This is due to the mass ratio being 3.33× in the underestimation case, compared to only 1.7× in the overestimation case. A more balanced selection of assumed masses would provide a more equitable basis for comparing the two scenarios. Furthermore, both cases suggests that the tuning of the parameters could have been more conservative to improve the robustness to model mismatch. Firstly, in the underestimation case, the induced overshoot and oscillations shows that less aggressive tuning would have been preferable. Lastly, in the overestimation case, the reduced RMSE in slope 10 also shows that a more conservative approach could lead to better performance.

In the Maggy system, the mass of the disc is perfectly known by the controllers. However, the system introduces other model mismatches as it is difficult to fully capture the complex system dynamics. As such, the robustness analysis is still insightful and transferable for the Maggy.

6.2 Future work

This section describes the work areas for future work based on the limitations and results of this project. The work areas are divided into short-term, medium-term and long-term prospects.

6.2.1 Short-term

- **Implementation of Δu_{max}**: The Stochastic and Tube MPC do not currently include constraints on the maximum change in control force between two consecutive sampling intervals. This should be implemented like in the Standard MPC case, so that these controllers can be objectively compared against the other controllers in terms of tracking performance.

6.2.2 Medium-term

- **Implementation on the Maggy simulator**: The work done has primarily focused on the simplified 1-DOF system. Future work should implement the `acados` MPC controllers on the Maggy simulator. Note that the simulator is written in `MATLAB`, while the simulator for the 1-DOF system is written in `Python`. Make a unified code base for the simulator with all the different controllers and estimators, which focuses on modularity and user-friendliness.

6.2.3 Long-term

- **Hardware Deployment**: Implement the `acados` MPC controllers on the physical Maggy. Investigate the possibilities for "magrobatics", which is more complex maneuvers than just maintaining levitation, such as circular movements and movement upwards and downwards. Examine the possibility

to integrate an event-based camera to the system to drastically improve the computational efficiency of the estimator.

CHAPTER

SEVEN

CONCLUSION

This thesis evaluated advanced Model Predictive Control (MPC) strategies for MagLev systems using a 1-DOF benchmark. By deploying Standard, Stochastic, and Tube-based architectures within the `acados` framework, the research demonstrated that robust, high-performance control is achievable within stringent real-time constraints. Compared to a `CasADi` baseline, `acados` provided a significant computational advantage, achieving speedups of 22.3× for short prediction horizons and 7.6× for longer horizons. Crucially, this efficiency did not compromise control quality, as `acados`-based controllers yielded comparable tracking accuracy and the lowest overall RMSE.

Systematic parameter exploration via Response Surface Methodology (RSM) revealed that tracking performance is primarily driven by the interaction between height and velocity weighting rather than horizon length. This suggests that computational loads can be further optimized using shorter horizons without meaningful precision loss. Additionally, robustness analysis showed that while the controllers are sensitive to mass underestimation, they exhibit strong resilience to overestimation, a finding that underscores the necessity of conservative tuning in physical applications.

Ultimately, the 1-DOF implementation serves as a proof of concept for the complex Maggy system. The developed code base and estimator designs are highly transferable; despite increased dimensionality, the underlying control loop utilizing the `acados` OCP solver remains identical. By providing a verified, real-time feasible implementation, this thesis lays the groundwork for future work, where the developed tools can be deployed to stabilize the full-scale Maggy platform.

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